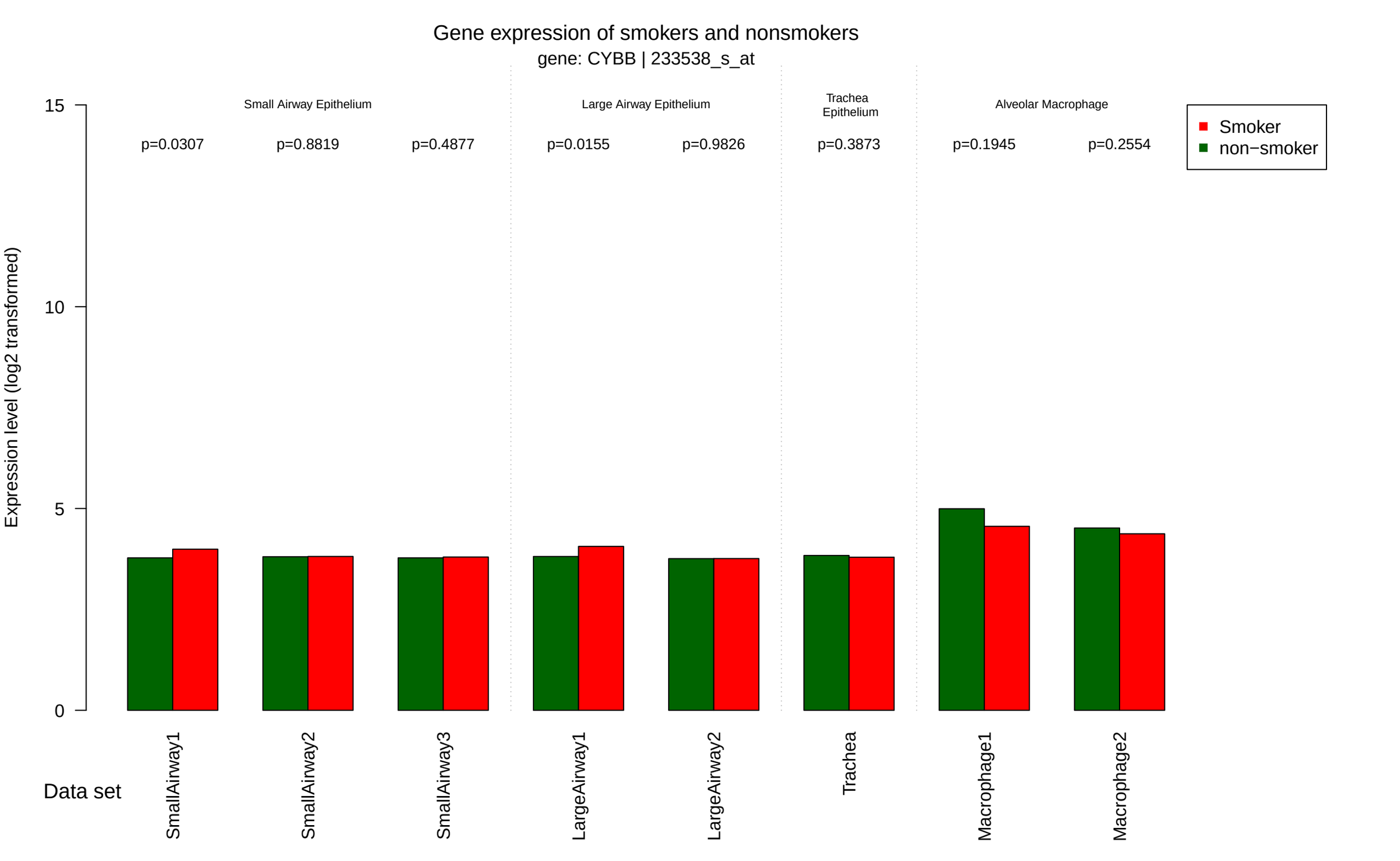
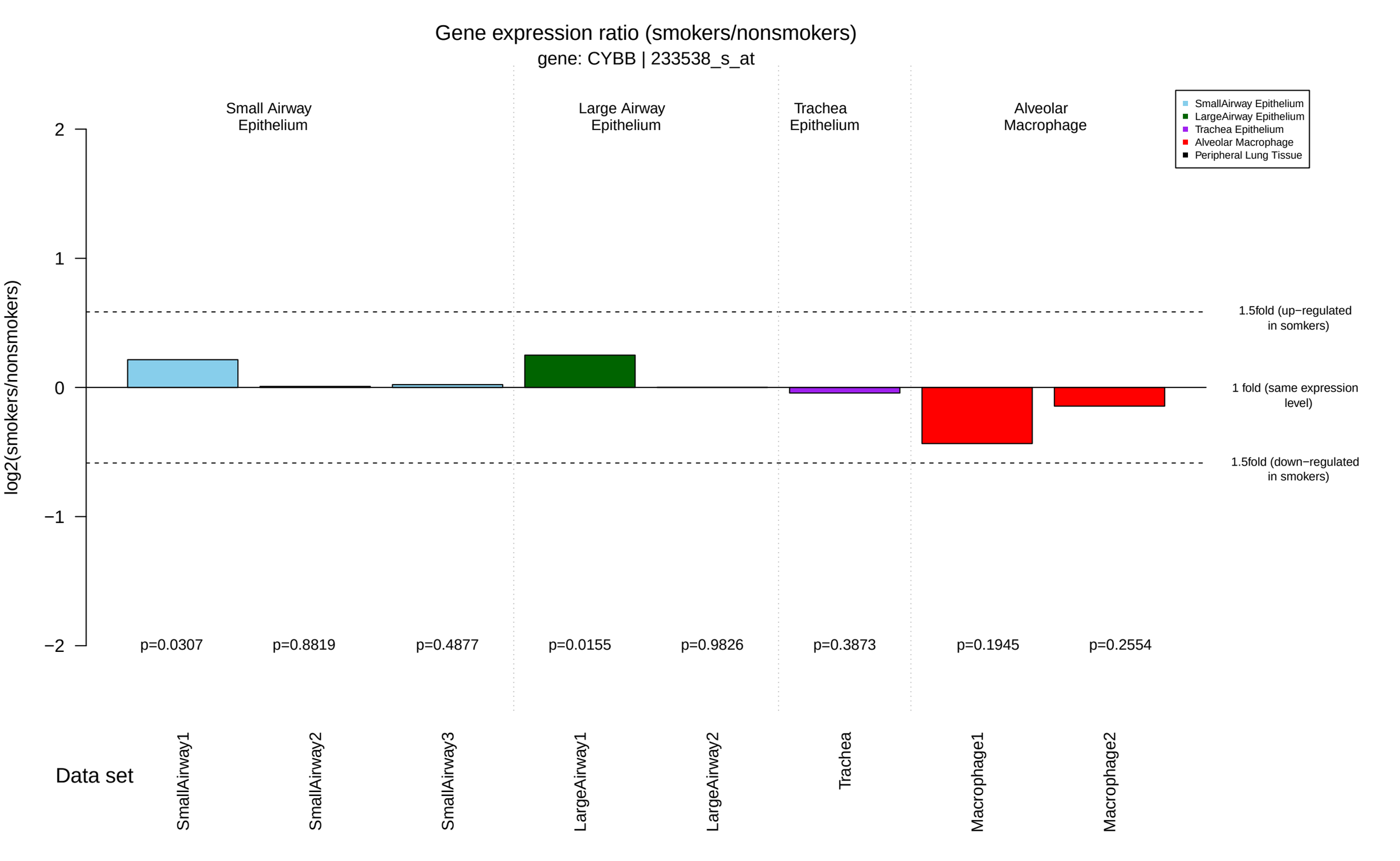
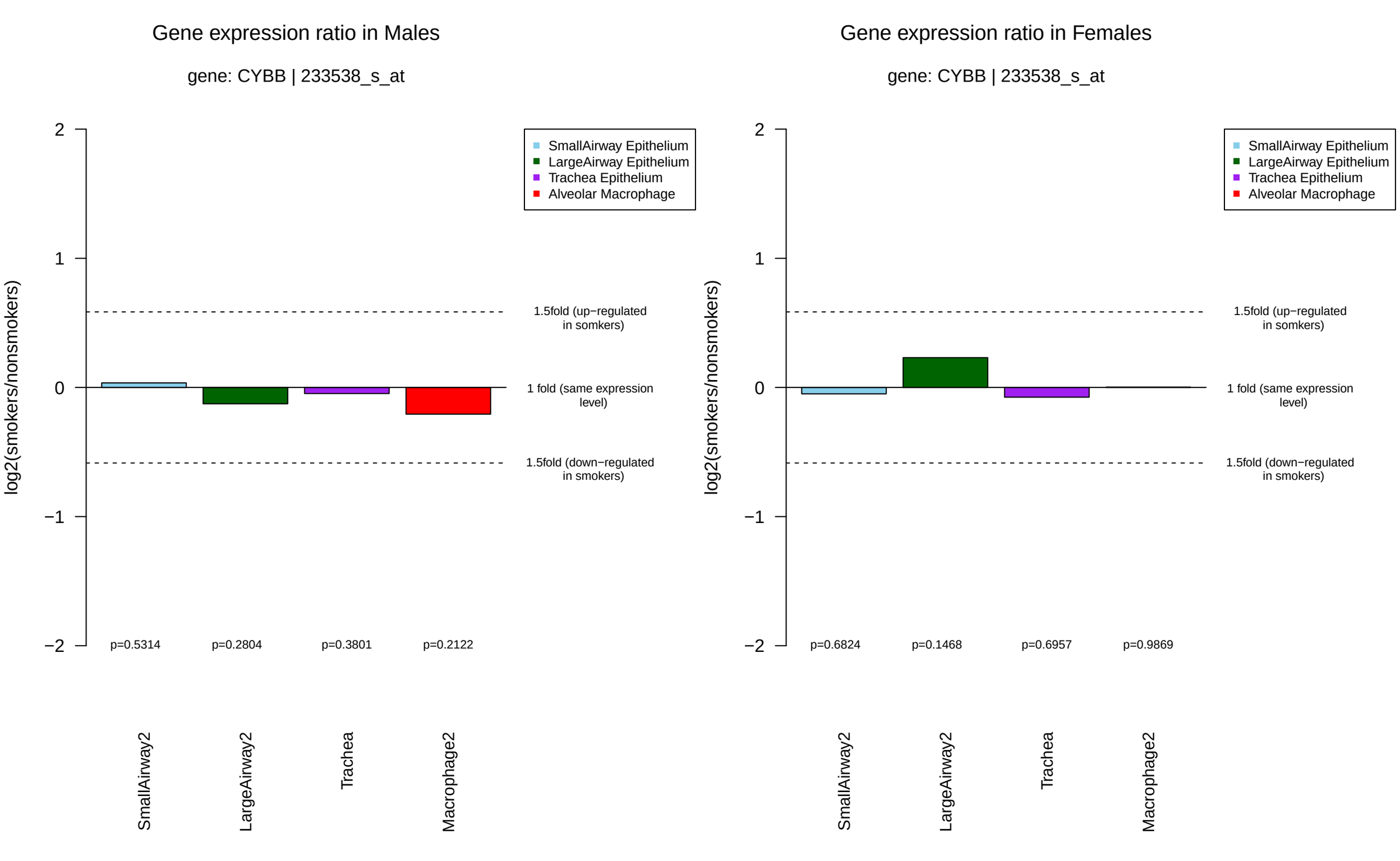
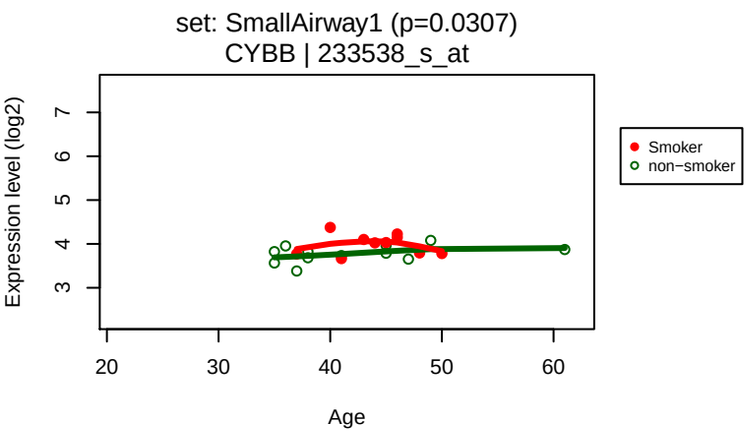
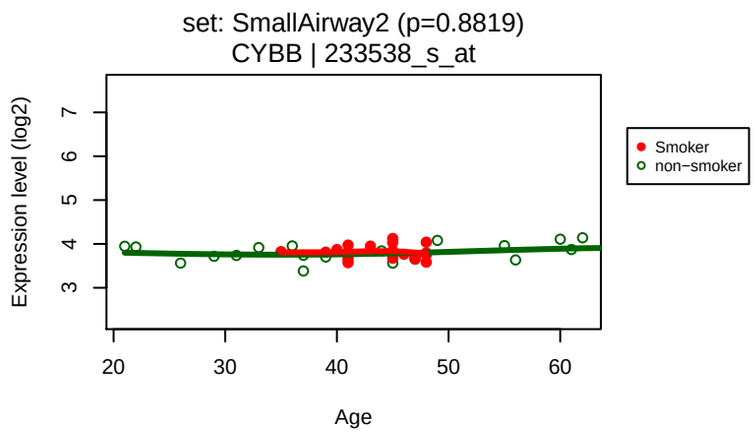
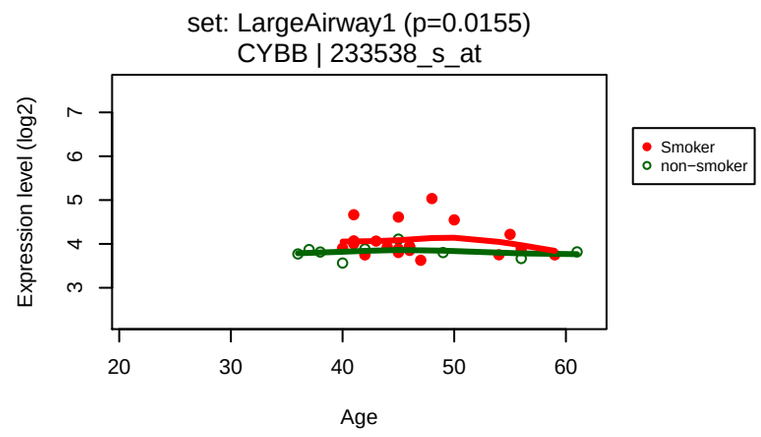

Gene expression of smokers and nonsmokers
gene: CYBB | 233538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0307
p=0.8819
p=0.4877
p=0.0155
p=0.9826
p=0.3873
p=0.1945
p=0.2554
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

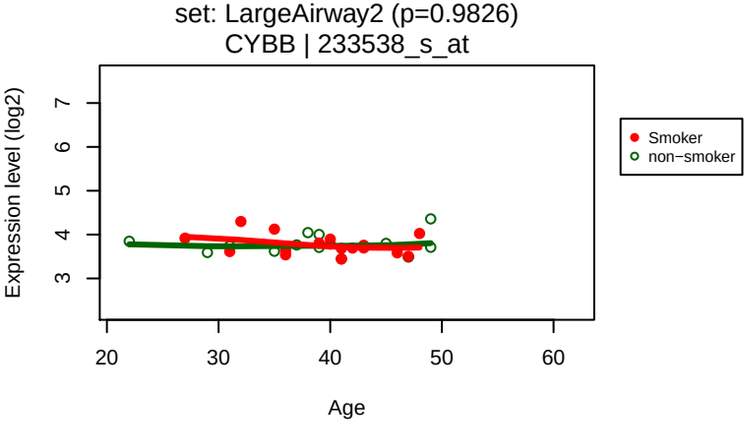

Gene expression ratio (smokers/nonsmokers)
gene: CYBB | 233538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0307
p=0.8819
p=0.4877
p=0.0155
p=0.9826
p=0.3873
p=0.1945
p=0.2554
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

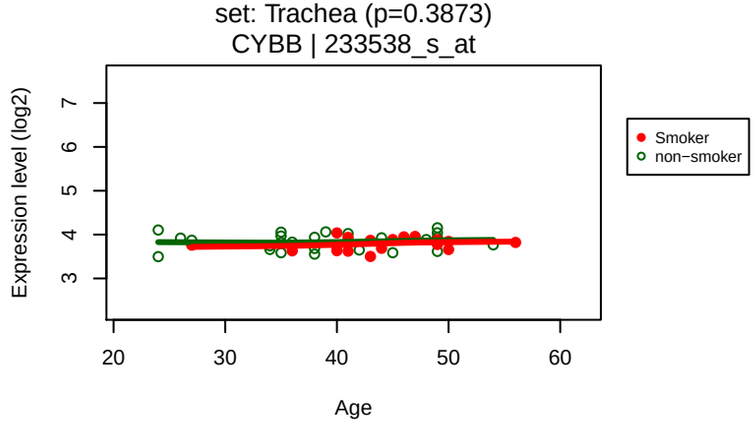

Gene expression ratio in Males
gene: CYBB | 233538_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5314
p=0.2804
p=0.3801
p=0.2122
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYBB | 233538_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6824
p=0.1468
p=0.6957
p=0.9869
SmallAirway2
LargeAirway2
Trachea
Macrophage2

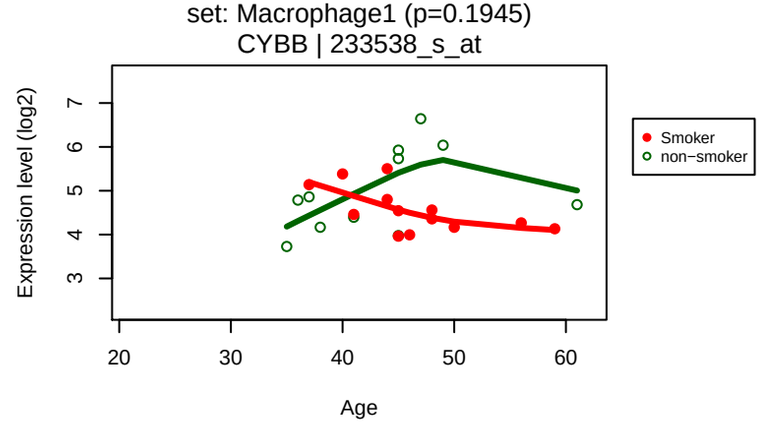

set: SmallAirway1 (p=0.0307)
CYBB | 233538_s_at
Expression level (log2)
Age
Smoker
non−smoker

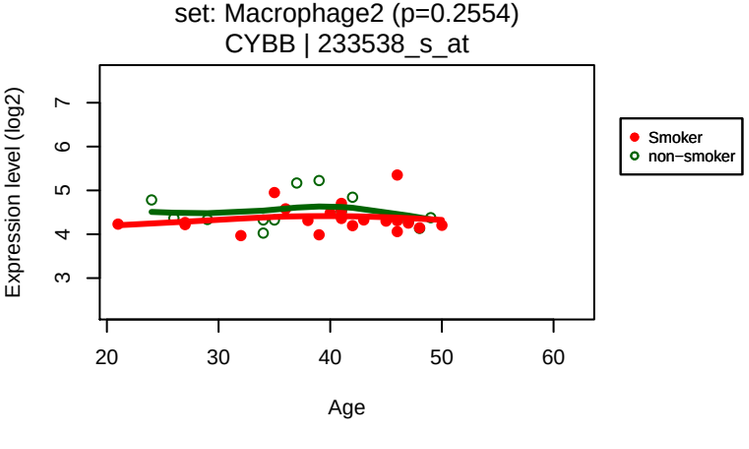

set: SmallAirway2 (p=0.8819)
CYBB | 233538_s_at
Expression level (log2)
Age
Smoker
non−smoker

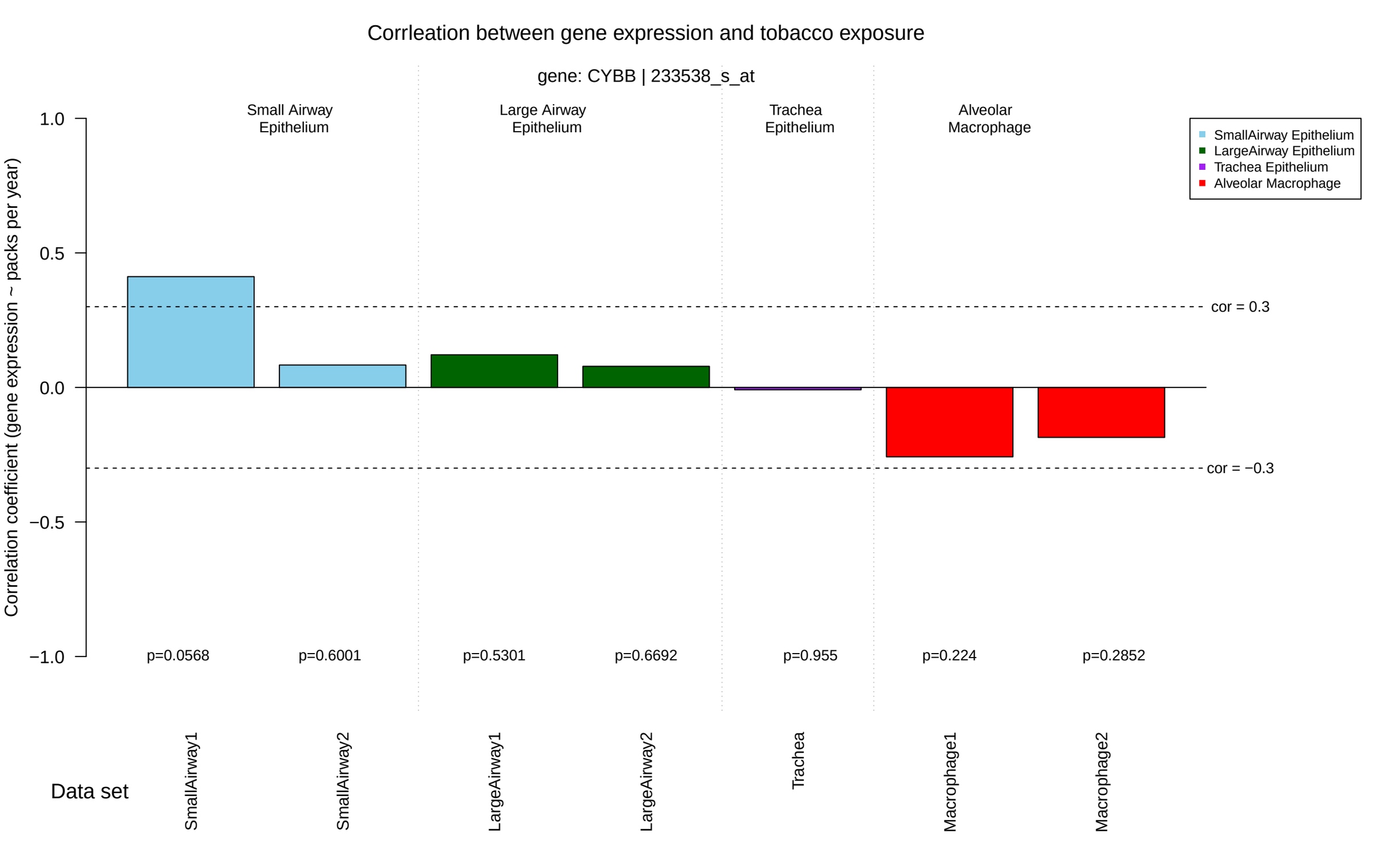

set: LargeAirway1 (p=0.0155)
CYBB | 233538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9826)
CYBB | 233538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3873)
CYBB | 233538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1945)
CYBB | 233538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2554)
CYBB | 233538_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYBB | 233538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0568
p=0.6001
p=0.5301
p=0.6692
p=0.955
p=0.224
p=0.2852
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2